Supplement: PRISM CPU/GPU Portfolio Optimization

Eligibility Ledger, Scale Frontier, and Workflow Sanity Checks

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This supplement accompanies *PRISM: A CPU/GPU Portfolio Optimization Engine for Deadline-Bounded Institutional Rebalancing*. It records material that supports the main paper without being required for its central claims: the solver eligibility ledger, the CPU/GPU scale frontier, workflow sanity checks, and additional artifact details. All evidence-artifact names, timing lanes, status vocabulary, and claim rules are those defined in the main paper.

A Solver Eligibility Ledger

Table S1 records the full eligibility ledger for the evidence package, including installed packages that did not produce claim-bearing rows. Entries are operational eligibility records under the declared protocol, not solver critiques.

Table S1. Solver eligibility ledger for the recorded evidence package.

Solver stack	Version	Eligibility
PRISM-CPU	evidence build `512758f`	claim-bearing where recorded
PRISM-GPU	evidence build `512758f`	claim-bearing; `e6` rows are timing/status only
OSQP	1.1.0	claim-bearing where recorded
Clarabel	0.11.1	claim-bearing where recorded
SCS	via CVXPY 1.8.1	claim-bearing where recorded
MOSEK	11.1.11	claim-bearing where recorded
CVXPY interface	1.8.1	comparator modeling layer; backend named per row
Commercial reference (anonymized)	recorded in evidence	objective-gap reference only; timings unpublished under license terms
CPLEX	22.1.2.0	configured license did not support rows at $N \geq 500$; completed public-small lane only; not eligible for claim-bearing comparisons
HiGHS	1.13.0	public-small lane only; not a QP comparator in this package
NVIDIA cuOpt	26.2.0	no versioned active run recorded; not benchmarked
CVXPortfolio	not installed	not benchmarked

B Scale Frontier Figure and Discussion

Table S2 and Figure S1 record the full-call CPU/GPU scale frontier from evidence `e1_multisolver_public_rows` and `e4_deadline_completion_grid`. Full-call timing includes public-call setup, data movement, status handling, validation, and output materialization; the frontier is a systems-boundary measurement, not a device-throughput claim. The convergence of CPU and GPU wall-clock values at selected sizes is a public-call systems effect, not evidence about device-level equivalence.

Table S2. PRISM CPU/GPU scale frontier, full-call wall-clock, evidence `e1/e4`. Rows without completed reference solvers carry timing and feasibility status only.

N	CPU full-call ms	GPU full-call ms	Reading (systems lane)
5,000	1,804.8	**140.7**	GPU path is the fastest recorded PRISM path on this row.
10,000	228.3	**215.9**	Both PRISM paths complete inside 0.5 s.
25,000	1,063.6	**458.3**	GPU path remains inside 1 s.
50,000	2,187.9	**861.4**	GPU path remains inside the 5 s deadline lane.
100,004	**2,157.8**	2,161.2	Public-call systems effect; see text.

Claim notes. `e4` also records GPU memory below 100 MB at N=100,004, treated as an externally observed memory-class result.

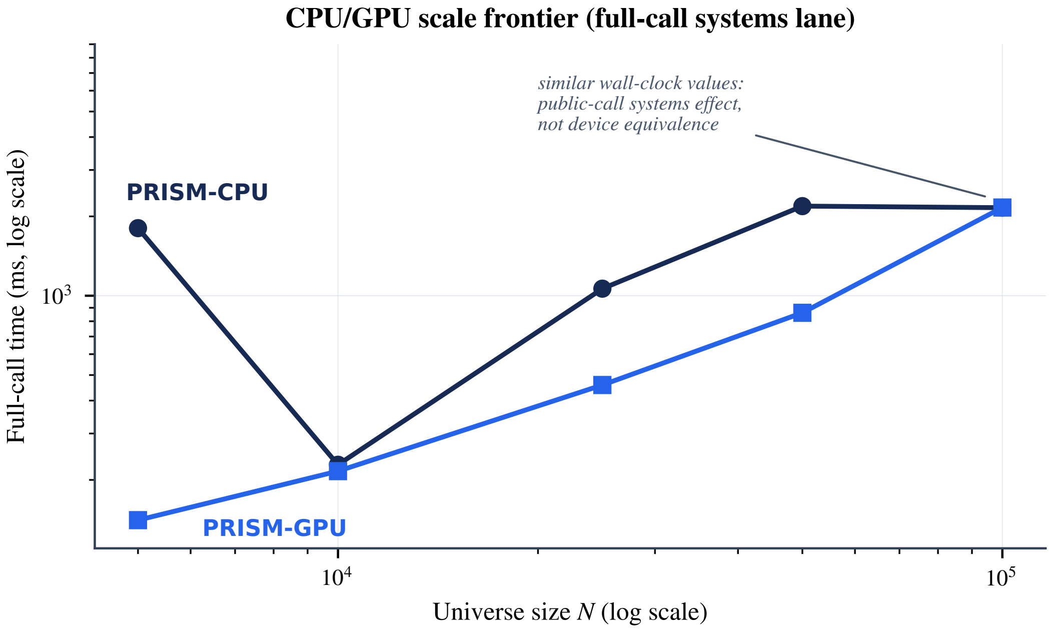

Figure S1. CPU/GPU scale frontier from Table S2, evidence `e1/e4`. Full-call timing includes public-call setup, data movement, status handling, validation, and output materialization; the figure is a systems-boundary measurement, not a device-throughput claim.

The deadline-completion grid in `e4` complements the frontier: it records, for each size and declared deadline from 0.1 s to 300 s, whether each PRISM path completed inside the window. PRISM-GPU completes the N=10,000 row inside 0.5 s and the N=100,004 row inside 10 s in that grid; both PRISM paths complete every recorded size at the 30 s and 300 s deadlines.

C Walk-Forward and Monte Carlo Sanity Checks

These checks verify pipeline behavior on repeated portfolio outputs. They are workflow-correctness evidence under the main paper's claim contract and are not investment-performance evidence.

C.1 Walk-Forward Check

The walk-forward check in `e3_external_diagnostics_and_gap_checks` covers 131 U.S. public-equity survivors from January 3, 2019 through December 31, 2025, with 300 weekly rebalance points, 5 bps transaction costs, and a 5% position cap. PRISM-Regime reports Sharpe 0.901 with 95% interval [0.770, 1.033], Sortino 1.144, and maximum drawdown −36.4%. Equal-Weight reports the same Sharpe in this check; Min-Vol reports 0.923. Two disclosures bound the interpretation. First, the universe is survivorship-biased by construction and the rolling-mean return input is non-predictive; the study is a solver and workflow correctness check. Second, the tie with Equal-Weight is expected: under a non-informative return input, converging to broad diversification is the mathematically correct outcome.

C.2 Monte Carlo Check

The Monte Carlo study in e3 evaluates 1,000 random universes of 20 to 50 stocks. PRISM reports mean Sharpe 5.315 with positive Sharpe in 79.2% of trials; Equal-Weight reports 3.974 and 75.1%; Min-Vol reports 5.095 and 84.0%. PRISM beats Equal-Weight in 60.4% of trials and Min-Vol in 54.1%. Cross-trial dispersion is large (standard deviation above 6 Sharpe units), which is why these numbers support workflow stability rather than any performance ranking.

D Additional Artifact Details

The public artifact repository (`AsymmetryComputing/prism-public-evaluation`) uses the following layout. PRISM implementation code is not part of the public artifact.

Table S3. Public artifact repository layout.

Path	Contents
`README.md`, `MANIFEST.yml`, `CITATION.cff`, `LICENSE`	Index, artifact manifest, citation metadata, license.
`environment/`	Hardware/runtime disclosure and package versions.
`data_sanitized/`	Benchmark input manifest; no client data, no engine state.
`results/`	One sanitized CSV per evidence artifact (`e1`–`e6`).
`figures/`	Published figures (vector).
`tables/`	Machine-readable copies of the published tables.
`validation/`	Public residual and feasibility-gate scripts; table reproduction script.
`evidence_ledger/`	Claim-to-evidence ledger, artifact manifest, artifact policy.
`paper/`, `supplement/`	arXiv version and sources.

Each claim-bearing number in the main paper maps to a row in `evidence_ledger/claim_to_evidence_ledger.md`, which records claim ID, claim text, section, table or figure, evidence artifact, allowed claim type, disallowed interpretation, and reproducibility level.

E Extra Tables and Claim Notes

Small-problem objective agreement (`e2_small_problem_agreement`). On a public FF30 small-problem study (five seeds, $N=30$), all completed comparators agree on the objective value to approximately seven significant figures, anchoring the objective-agreement methodology used for gap eligibility. PRISM timing rows are not recorded in `e2`; the artifact is comparator-agreement evidence only.

Status vocabulary. The badges used in the main paper's tables abbreviate the declared status vocabulary: *feas* (feasible / non-certified), *t/o* (timeout at the declared ceiling), and *late* (completed status after the operating window). "No accepted feasible return" records that no output passed the external feasibility gates before the deadline; it is not a certified infeasibility status.

Pending evidence. A burden-specific PRISM-GPU rerun of the constraint-burden curve and a multi-seed rerun of the single-shot `e1` rows are recorded as planned evidence extensions.